

# BeamStellar: Low-Latency, 6-DoF Head Tracking for Beaming Displays with Spatio-Temporal LED Encoding

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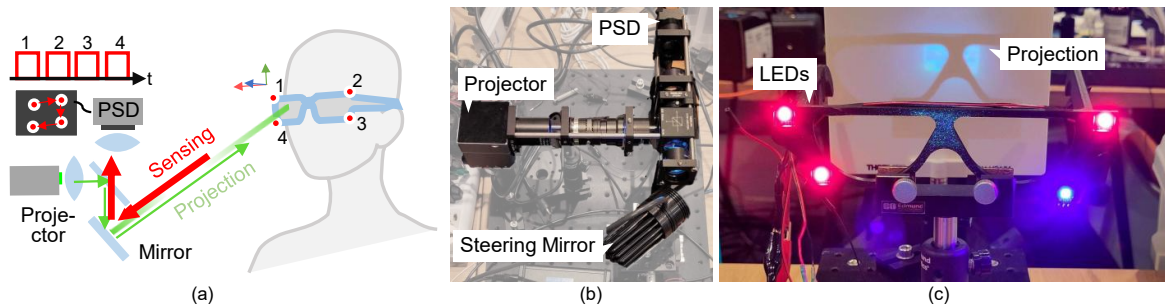


Figure 1: An overview of our BeamStellar concept, a low-latency 6DoF tracking design for beaming display systems. (a) a schematic diagram of the projection-sensing unit with a PSD (position sensing detector) and the light-receiving glasses with a coded LED array. (b) A benchtop system of the projection unit. (c) a mockup of glasses with an LED array. The colors of LED are irrelevant to the algorithm. A projected light from the projection-sensing unit is visible on a screen behind the glasses.

## ABSTRACT

Conventional augmented reality (AR) head-mounted displays (HMDs) require the integration of all components, including optics and processors, which presents significant barriers to practical implementation due to weight and heat generation concerns. While Beaming Displays address these challenges by offloading computing and projection functions to the environment, existing solutions leaving 6-degrees of freedom (6DoF) image projection with low motion-to-photon (M2P) latency. In this position paper, we introduce our trial BeamStellar, a Beaming Display system that integrates 6 DoF head tracking by combining spatio-temporally encoded LED patterns with a position-sensing detector. Through digital signal processing using an FPGA, the system aims a theoretical latency of less than 200 microseconds. This implementation alleviates the hardware burden on users, facilitating seamless AR experiences that seamlessly blend virtual content with the physical world.

**Keywords:** Beaming Displays, Optical See-through Display, Low-latency head tracking, Spatio-temporal Light pattern

## 1 INTRODUCTION

Optical see-through head-mounted displays (OST-HMDs) overlay digital content directly into the user's field of view (FoV), seamlessly integrating virtual and physical environments. Despite their potential as a key technology for augmented reality (AR) experiences, conventional OST-HMDs face significant trade-offs among critical performance factors such as weight, power consumption, and computational capacity. The integration of essential components, including optics, sensors, batteries, and processors, into a compact form factor often results in compromises, such as reduced battery life, increased heat load, and limited FoV and computing power.

To overcome these challenges, Itoh et al. [5] proposed the Beaming Display approach, which offloads computational and optical processing to the surrounding environment. In this concept, projection units are placed externally, while users wear lightweight,

passive headsets to view AR images. This reduces the user's hardware burden while enabling high computational power. However, practical implementation of Beaming Display systems requires addressing a critical technical hurdle: achieving low motion-to-photon (M2P) latency [6] while tracking freely moving viewpoints. While Hiroi et al. [4] demonstrated a beaming display system with a latency of 133  $\mu$ s using a 2D position-sensing detector (PSD) and a single LED, their approach was limited to tracking 3D position (i.e., xyz) only, lacking the full 6 degrees-of-freedom (DoF) pose estimation necessary for dynamic AR environments.

In this position paper, we present BeamStellar, an ongoing project to achieve full 6-DoF headset tracking in Beaming Display systems. Our approach leverages spatiotemporally encoded LED patterns and a PSD to ensure precise alignment between projected images and headset position. By integrating digital signal processing on an FPGA, our system can minimize processing latency while maintaining accurate headset pose tracking. This work outlines BeamStellar's design principles with current hardware design and analyzes the architecture's theoretical latency performance.

## 2 BEAMSTELLAR CONCEPT

BeamStellar's goal is to achieve low-latency 6 DoF headset tracking in Beaming Displays by combining spatio-temporally encoded LED patterns with a single PSD.

### 2.1 System Architecture

Figure 1 shows an overview of the BeamStellar system architecture. The system consists of two main units: a projection-sensing module and light-receiving glasses with an LED array (Fig. 1(b)). The tracking works as follows: the LEDs on the glasses turn on sequentially, and then the module captures each LED and detects their 2D positions, allowing the reconstruction of the headset's 6DoF pose (Fig. 1(a)).

The projection-sensing module has a PSD, steering mirror, and projector. The sensing and the projection paths align optically to keep the tracking and projection in the same FoV. A PSD is an analog 2D sensor that can precisely detect the center of projected light at a fast analog response. For example, our PSD (Thorlabs 2D Lateral Effect Position Sensor PDP90A) has a position resolution

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## REFERENCES

- [1] K. Akşit and Y. Itoh. Holobeam: Paper-thin near-eye displays. In *Proc. of IEEE VR 2023*, pages 581–591. IEEE, 2023. [2](#)
- [2] H. Aoki, T. Tochimoto, Y. Hiroi, and Y. Itoh. Towards co-operative beaming displays: Dual steering projectors for extended projection volume and head orientation range. *IEEE TVCG*, 30(5):2309–2318, May 2024. [2](#)
- [3] A. Blate, M. Whitton, M. Singh, G. Welch, A. State, T. Whitted, and H. Fuchs. Implementation and evaluation of a 50 khz, 28us motion-to-pose latency head tracking instrument. *IEEE TVCG*, 25(5):1970–1980, 2019. [2](#)
- [4] Y. Hiroi, A. Watanabe, Y. Mikawa, and Y. Itoh. Low-latency beaming display: Implementation of wearable, 133  $\mu$ s motion-to-photon latency near-eye display. *IEEE TVCG*, 29(11):4761–4771, Nov. 2023. [1](#), [2](#)
- [5] Y. Itoh, T. Kaminokado, and K. Akşit. Beaming displays. *IEEE TVCG*, 27(5):2659–2668, 2021. [1](#)
- [6] P. Lincoln, A. Blate, M. Singh, T. Whitted, A. State, A. Lastra, and H. Fuchs. From motion to photons in 80 microseconds: Towards minimal latency for virtual and augmented reality. *IEEE TVCG*, 22(4):1367–1376, 2016. [1](#)